

WASSWA WILSON

Biomedical Engineer · Full-Stack Software Developer · Health Technology Programme Lead

Kampala, Uganda · Available for regional and remote engagements
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English (fluent, professional working proficiency) · Luganda (native)

PROFESSIONAL PROFILE

Biomedical engineer and full-stack software developer with over nine years of professional experience spanning hospital technology, health information systems and applied software engineering. I currently lead development of the FairBanks Community Health Intelligence Platform (FCHIP), owning the data capture tools, database design, cloud data flows and management dashboards end to end. Before that I managed the Gould Family Foundation biomedical programme across East and Central Africa and ran the biomedical function at International Hospital Kampala through successful COHSASA accreditation. I build in Java, Python, JavaScript and TypeScript across React, Next.js, React Native, Node.js and Express, and I design AI agents and automation pipelines that are running in production today. I am equally comfortable commissioning equipment on a hospital floor, architecting a codebase, and writing the funding proposal that pays for both.

SIGNATURE ACHIEVEMENTS

- Lead developer of FCHIP, the FairBanks Community Health Intelligence Platform — community screening, referral and follow-up data capture, database, cloud sync and reporting dashboards, delivered across web and mobile.
- Designed, built and deployed an AI reception agent for a law firm in Texas, USA — it handles live inbound client calls, books and reschedules appointments, and runs client follow-up. In active production use.
- Delivered hospital technology projects in five countries — Uganda, the Democratic Republic of Congo, Kenya, Tanzania and Somaliland — including laboratory, theatre and radiology installations at the hospital serving the Kibali Gold Mine, DR Congo.
- Prepared and maintained the biomedical compliance evidence that carried International Hospital Kampala through successful COHSASA accreditation, and cut equipment downtime with a documented preventive maintenance programme.
- Upgraded partner health facilities across East and Central Africa for the Gould Family Foundation, including the neonatal intensive care unit at Mama Lucy Kibaki Hospital, Nairobi.
- Author of institutional proposals, grant applications and partnership documents for FairBanks Medical Centre; chaired the FairBanks / Uganda Health Partners Cooperative meeting on cooperative health insurance in July 2026.

TECHNICAL SKILLS

Programming languages	Java · Python · JavaScript · TypeScript · C · C++ · C# · PHP · SQL · MATLAB · Arduino C · HTML5 · CSS3
Web & mobile	React · Next.js (App Router) · React Native · Expo · Tailwind CSS · styled-components · Bootstrap · JavaFX / FXML · Jest · Playwright
Backend & APIs	Node.js · Express · Laravel · CodeIgniter · Spring / Spring Boot · Convex · Prisma · REST APIs · JWT · role-based access control
Databases & data	MySQL · relational modelling and query optimisation · SQL analytics · data cleaning and reporting · MATLAB modelling · Excel analysis and dashboards
Cloud & DevOps	AWS · Microsoft Azure · Vercel · EAS · Bunny.net CDN · Git and GitHub · CI/CD pipelines · release and environment management
AI, agents & automation	AI agent design, deployment and monitoring · workflow and process automation · third-party systems integration · automated auditing and verification tooling · Cursor · Claude · OpenAI GPT · Codex
Operating systems	Windows · Linux · Android — administration, deployment and troubleshooting
Engineering & scientific	Embedded systems design · biosignal processing · medical imaging · biomedical modelling and simulation · bioinformatics and functional genomics · Proteus · Eagle CAD · Solid Edge
Documents & productivity	Microsoft Word · Excel · PowerPoint · technical writing · proposal and grant development · document editing and design · report and dashboard production

PROFESSIONAL EXPERIENCE

Lead Software Developer — FCHIP and Health Information Systems

2026 – Present

FairBanks Medical Centre Limited, Kampala, Uganda

- Lead the design and build of the FairBanks Community Health Intelligence Platform (FCHIP): data capture form design, database structure, cloud data flows, reporting dashboards, and delivery across both web and mobile.
- Architect and maintain the hospital and clinic software used daily by clinical and administrative staff, covering patient records, scheduling, referral tracking and management reporting.
- Design and deploy AI agents and automation pipelines that remove manual handling from reception, appointment management, client follow-up and routine reporting.
- Own technical documentation, data-protection practice, access control and release management across the platform.
- Write and edit institutional proposals, grant applications, partnership documents and investment materials; chaired the FairBanks meeting with the Uganda Health Partners Cooperative on cooperative health insurance in July 2026.

Consultant Biomedical Engineer — Independent Practice

Mar 2025 – Present

Contract assignments: Uganda, DR Congo, Kenya, Tanzania, Somaliland

- Install, commission, calibrate and maintain hospital medical equipment under contract across five countries, working independently on site and to a fixed handover standard.
- Completed laboratory, theatre and radiology installations at several hospitals in the Democratic Republic of Congo, including the facility serving the Kibali Gold Mine.
- Train clinical users at handover and produce a written service report and maintenance record for every assignment, so the client keeps a verifiable equipment history.
- Handle preventive and corrective maintenance, on-site fault diagnosis and specification advice for procurement.

Biomedical Programs Manager

Aug 2024 – Feb 2025

Gould Family Foundation (GFF) — East and Central Africa

- Managed the regional biomedical team, covering programme planning, implementation and reporting across partner health facilities in multiple countries.
- Oversaw procurement, installation and commissioning of medical technologies, with programme documentation kept in line with healthcare technology standards.
- Developed equipment lifecycle-management and master-planning strategies that were applied across partner facilities.
- Led biomedical and clinical user training programmes for engineers, technicians and clinical staff.
- Upgraded several facilities, including the neonatal intensive care unit at Mama Lucy Kibaki Hospital, Nairobi.

Biomedical Manager

Jan 2020 – Jan 2024

International Hospital Kampala (IHK)

- Managed the biomedical engineering function of a major private hospital for four years, including team supervision, workplans and reporting to executive management.
- Led installation and commissioning of CT and X-ray imaging, oxygen plant, laboratory, theatre and ICU systems, through to handover and trained clinical users.
- Built the preventive maintenance schedules and service-record systems that reduced equipment downtime and kept devices safe for clinical use.
- Coordinated procurement, technical specification, commissioning documentation and lifecycle management of hospital medical equipment.
- Prepared and maintained the compliance evidence files reviewed for COHSASA accreditation, supporting the hospital through successful accreditation.

Biomedical Engineer

Apr 2019 – Jan 2020

Norvik Hospital Ltd, Kampala

- Maintained and serviced hospital diagnostic and patient-monitoring equipment.
- Supported installation and calibration of medical imaging and ICU equipment.
- Helped establish preventive maintenance schedules and service follow-up records.

Research Intern

Dec 2018 – Apr 2019

Uganda Virus Research Institute (UVRI)

- Contributed to biomedical research projects and laboratory experimentation.
- Carried out laboratory data analysis, mathematical modelling and research pipeline design in Python.

Teaching Assistant

Mar 2016 – Aug 2017

Makerere University, College of Health Sciences

- Assisted in teaching biomedical engineering course units at undergraduate level.
- Supervised and supported laboratory practical sessions for engineering students.

SELECTED PROJECTS AND SYSTEMS DELIVERED

FairBanks Community Health Intelligence Platform (FCHIP)

2026 – present

Lead Software Developer — FairBanks Medical Centre Limited

Community, screening, referral and follow-up data capture; database structure; cloud data flows; management dashboards; web and mobile delivery. Built with Next.js, TypeScript, React Native and cloud data services.

AI reception agent

2025 – present

Designer and Developer — Law firm, Texas, USA

Built and deployed an AI agent that answers live inbound client calls, books and reschedules appointments, and runs post-contact client follow-up. In active production use.

Hospital equipment installation and commissioning programme

2020 – 2024

Biomedical Manager — International Hospital Kampala

CT, X-ray, oxygen plant, laboratory, theatre and ICU systems. Led installation, commissioning and handover, with preventive maintenance and service documentation that cut downtime.

COHSASA hospital accreditation programme

2020 – 2024

Biomedical Manager — International Hospital Kampala

Maintained equipment, maintenance and compliance evidence to accreditation standard and supported the hospital through successful accreditation.

Shulekeeper school information system

2021 – 2024

Assistant Software Developer — Shulekeeper

Multi-user records and reporting platform built with React and Next.js, TypeScript, Laravel, MySQL and AWS. Contributed front-end and back-end work, third-party integrations, testing, user training and documentation.

Endoscopy image capture and reporting software

2017 – 2019

Lead Software Developer — St. Catherine Hospital and St. Francis Hospital

Clinical image capture and procedure reporting system in JavaFX and MySQL. Led algorithm design, relational modelling, legacy data migration, on-site installation, user training and the user manual.

Wekebere foetal heart-rate monitoring system

2017 – 2024

Electronics Developer and Programmer — Wekebere

Android and Arduino monitoring system for third-trimester foetal heart-rate measurement, with Java serial communication between device and phone and Microsoft Azure cloud storage.

PROFESSIONAL COMPETENCIES

Programme and project management	Managed a regional biomedical team across East and Central Africa and a hospital biomedical function for four years: workplans, schedules, procurement coordination and management reporting.
Monitoring, evaluation and data	Leads the FCHIP indicator, data-quality, dashboard and reporting build. Built equipment registers, maintenance schedules and routine management reports. Analysis in Python, SQL, Java, JavaScript and MATLAB.
Proposal, grant and technical writing	Writes and edits institutional proposals, grant applications, investment materials, partnership documents, technical documentation and user manuals.
Data analysis and automated auditing	Cleans, models and reports on operational and clinical data; builds automated auditing and verification tooling to check records at scale rather than by hand.
Procurement, specification and compliance	Coordinated specification, procurement, commissioning and lifecycle documentation, and kept the compliance evidence reviewed for COHSASA accreditation.
Stakeholder and partnership management	Chairs partner meetings and writes them up; works with hospital departments, suppliers, trainees, accreditation assessors, donors and prospective partners.
Training and capacity building	Trained biomedical engineers, technicians and clinical users across multiple facilities, and taught undergraduate engineering laboratory practicals at Makerere University.

Communication and adaptability

Clear written and spoken English, comfortable presenting to clinical, executive and donor audiences; settles quickly into new countries, teams, codebases and equipment brands.

MEDICAL EQUIPMENT AND CLINICAL TECHNOLOGY EXPERIENCE

Radiology and imaging	CT scanners · X-ray machines · ultrasound systems · C-arms
Theatre and intensive care	Theatre systems · ventilators · patient monitors · infusion pumps · critical-care devices
Laboratory	Haematology analysers · clinical chemistry analysers · blood bank and transfusion equipment
Maternity and neonatal	Neonatal incubators · phototherapy units · foetal Dopplers · CTG machines
Support plant	Oxygen plants · RO water treatment systems · medical gas distribution

EDUCATION

Certificate in Leadership and Management in Health

2022

University of Washington

Grade A+. Planning, leadership and management of health programmes.

BSc Biomedical Engineering — Second Upper Honours

2012 – 2017

Makerere University, Kampala

Key course units: Information and Communication Technology, Computer OOP Principles, Software Engineering, Database Systems, Computer-Based Medical Systems, Embedded Systems, Biosignal Processing and Analysis, Medical Imaging, Biomedical Modelling and Simulation, Bioinformatics and Functional Genomics, Clinical Engineering, Research Methods, Principles of Management.

Advanced Java Programming — Level 1

2015 – 2016

Green Bridge School of Open Technologies

Grade A+.

Uganda Advanced Certificate of Education (UACE)

2011

Mengo Secondary School

Principal subjects: Mathematics, Physics, Biology, Chemistry.

Uganda Certificate of Education (UCE)

2009

St. John's Wakiso Secondary School

PROFESSIONAL TRAINING AND CERTIFICATION

- **COHSASA accreditation standards — in-service preparation and compliance** — International Hospital Kampala, 2020 – 2024
- **Fire Safety, Prevention, Firefighting and Emergency Scene Management** — Fire Technologies Limited (IHK), 2021

INTERESTS

Music — plays the saxophone and other instruments. · Film. · Physical fitness and regular exercise. · Travel and working in new environments, which the regional field work has made second nature.

REFEREES

Eng. Richard Ssejongo

Biomedical Engineer, St. Francis Hospital Nsambya

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